

● RS002 Week 13 — English Worksheet

Topic: Random Battles — `import random` · **Course:** Pokédex App · **Time:** about 45 minutes

This week your dex becomes a game. You `import random` to pick wild Pokémon out of nowhere, then compare stats with `if` / `elif` / `else` to decide a winner. Run it twice and you get two different battles — that is **randomness**.

1 · Recall

From memory:

Write a function that takes a dictionary `p` and prints its name and attack.

2 · Predict 🧠

Read each block. Before you run it, write what you think will happen.

```
import random

wild = ["꼬렛", "구구", "캐터피"]
print(random.choice(wild))
print(random.choice(wild))
```

Could the two printed names be different? Could they be the same? Why?


```
a = {"name": "피카츄", "attack": 55}
b = {"name": "이상해씨", "attack": 49}

if a["attack"] > b["attack"]:
    print(f"{a['name']} 승리!")
else:
    print(f"{b['name']} 승리!")
```

Which name is printed? Why?


```
import random

dex = [{"name": "피카츄"}, {"name": "파이리"}]
mine = random.choice(dex)
print(mine["name"])
```

`random.choice` here picks a whole dictionary. What kind of thing is `mine` , and what prints?

3 · Spot the Bug 🐛

Read what each block is **supposed** to do, fix it, then explain the fix.

Bug A — This should pick a random Pokémon. Right now `random` was never imported, so it crashes.

```
dex = ["피카츄", "이상해씨", "파이리"]
print(random.choice(dex))
```

Hint: you must bring the tool in before using it.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — A draw (equal attack) should print `무승부`. Right now equal stats wrongly count as a loss.

```
a = {"name": "피카츄", "attack": 50}
b = {"name": "파이리", "attack": 50}

if a["attack"] > b["attack"]:
    print(f"{a['name']} 승리!")
else:
    print(f"{b['name']} 승리!")
```

Hint: add an `elif` for `<` and an `else` for the tie.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — This should compare two Pokémon's stats. Right now it compares a Pokémon to its own name string and crashes.

```
import random
dex = [{"name": "피카츄", "attack": 55}, {"name": "파이리", "attack": 52}]

mine = random.choice(dex)
enemy = random.choice(dex)

if mine["attack"] > enemy:
    print("승리!")
```

Hint: compare `enemy["attack"]`, not `enemy` itself.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Modify It

Start from this working stat battle:

```
import random

pokedex = [
    {"name": "피카츄", "hp": 35, "attack": 55},
    {"name": "이상해씨", "hp": 45, "attack": 49},
    {"name": "파이리", "hp": 39, "attack": 52},
]

my_pokemon = random.choice(pokedex)
enemy = random.choice(pokedex)

print(f"\n 🍒 {my_pokemon['name']} vs {enemy['name']}")
print(f"   체력: {my_pokemon['hp']} vs {enemy['hp']}")

my_score = my_pokemon["hp"] + my_pokemon["attack"]
enemy_score = enemy["hp"] + enemy["attack"]

if my_score > enemy_score:
    print(f"\n 🏆 {my_pokemon['name']}의 승리!")
elif my_score < enemy_score:
    print(f"\n 😞 {enemy['name']}에게 졌습니다...")
else:
    print(f"\n ⚖️ 무승부!")
```

Make these changes one at a time and run after each:

1. Print each Pokémon's `attack` as well as its `hp` before the result.
2. Change the score formula to weight attack more (for example `attack * 2 + hp`).
3. Run the program five times and write down how many times you won.

Write your changed / added lines here, and your win count:

5 · Make It 📷

Add a **Battle!** option to your Week 12 menu. When chosen, it picks a random enemy from the dex and battles your Pokémon by comparing stats, printing the winner.

When it works, send a **video** on KakaoTalk showing the **battle run at least three times** so we can see the random results change. Then explain what you did. Use these starters — write 4 to 6 sentences.

First, I imported ... so I could ...

`random.choice` picked ... from my dex.

I decided the winner by comparing ...

Running it again gave a different result because ...

The trickiest part was ...

If I had more time, I would make the battle ...

6 · Record Your Walkthrough 🎥

Film several battles. Teach it like the viewer is new. Try to use: **import, random, choice, score, if / elif / else.**

1. Run the battle three times and point out the different match-ups.
2. Read your `random.choice` line out loud.
3. Explain how the winner is decided from the stats.
4. Say why the result changes each run.

Plan what you will say here first:

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.